

Mauro Alejandro Jimenez Medina

MSc Electronics Engineer and Sr Data Engineer

 Barranquilla, Colombia  mauroalejandrojim  jmauro@uninorte.edu.co

SKILLS

Machine Learning	NumpPy, Pandas, Scikit-Learn, Keras, PyTorch, NLTK, Weka, Google Colab.
Databases	MongoDB, PostgreSQL, MySQL, MariaDB, DynamoDB, SQLite, Turso.
Programming and Tools	Python3, Javascript (ES6), Node.js, Nuxt.js, Vue.js, C++, C#, VHDL, Spark, Arrow, Scala, Git, Docker, Google Cloud, AWS, Bitbucket, Agile (Scrum).
Languages	Spanish (Native) and English (ITEP).

EXPERIENCE PROFESSIONAL

DexFreight - Sr Data Engineer Barranquilla, Atlántico	08/2019 - Present
• Technical Leadership & Project Management: Led the end-to-end lifecycle of port operations optimization initiatives. Managed multidisciplinary teams and coordinated agile (Scrum) sprints, translating complex business requirements into data architectures and Machine Learning models, ensuring successful production deployment on azure cloud and reducing operational bottlenecks. • Deep Learning Models (NLP/Audio): Developed a VMRS decoding system employing a speech-to-text transformer model implemented with Python and PyTorch (Keras for logs), optimized to process audio samples with a signal-to-noise ratio (SNR) over -58dB. • Operations Optimization: Designed a fleet management algorithm based on state machines using Python, NumPy, and Pandas. This optimized trucking operations, reducing driver underutilization by 5%, deadhead miles by 27%, and non-revenue hours by 44%. • Routing & Prediction Algorithms: Built a trip planning algorithm combining geospatial features and Bayes' theorem via Python and Pandas, processing thousands of records with an average response time under 5 seconds. • Recommendation Systems: Designed and deployed a shipment recommendation model using Scikit-learn to match user preferences with historical attributes, achieving an average response latency of 600ms with multi-cloud support using docker images. • Developed robust ETL pipelines in Python for clustering geospatial data into map layers, optimizing data storage by 98% and improving the customer experience in the search system with support for GCP storage to use custom client artifacts.	
Blockchain Engineer - Technical Consultant Bogotá, Cundinamarca	01/2026 - Present
• Architecture design & proof of concept: Led the R&D of a proof of concept evaluating the implementation of distributed ledger technology (DLT) within a traditional banking ecosystem, focusing on secure ledger trust lists and real-world asset tokenization. • Technical & Strategic Documentation: Authored feasibility reports and presented technical findings to bank stakeholders, ensuring strict compliance with financial sector confidentiality agreements (NDA) while defining the infrastructure requirements needed to implement and scale the project.	
Chainomics ai - Technical Lead Engineer (LLMs) Barranquilla, Atlántico	01/2026 - Present
• Technical Leadership & Stakeholder Collaboration: Led the design and production deployment of LLM-based applications. Technical working sessions and executive reviews. • LLMs & Semantic Ranking: Designed and refined prompts for LLM-based semantic classification and developed robust similarity-based retrieval models. Diagnosed and resolved assignment accuracy issues by tuning retrieval thresholds and optimizing ranking behavior for high-cardinality matching.	
Psychology PhD Thesis Study - Data Analyst Barranquilla, Atlántico	06/2019 - 08/2019
• Designed and Analysis of experiments for an instrument that measures the falsehood or credibility of stories for children between 8 and 11 years old. • Data Analysis & NLP: Identified linguistic descriptors for the study group by applying Natural Language Processing (NLP) techniques using NLTK and Pandas, and extracted patterns based on neuropsychological knowledge using Machine Learning classification algorithms with Scikit-learn.	
MadnessLab - Backend Developer Barranquilla, Atlántico	06/2019 - 12/2019
• Machine Learning applied to QoS: Generated sensitive analytical reports for user and operator metrics, improving the quality of service by 20% through NLP and Machine Learning techniques implemented in Python and Keras/PyTorch. • ELT pipelines for processing call audio files recorded for credit institutions or financial companies with support for over 10 FFmpeg codecs.	

EXPERIENCE
ACADEMIC

Quatum Computing Framework - Python Developer	08/2021 - 10/2021
Remote, Mexico	
• Tutoring and subsidiary in Python and Numpy for the thesis modelling of a system/ environment in the framework of deeper entropy rise (SEA-QT) using random matrixes.	
Uninorte - Assistant Professor	07/2019 - 07/2020
Barranquilla, Atlántico	
• Instructed classes for digital signal processing using Matlab and Python focused on UI design using the Qt framework with an average score of 4.7/5 in the professor's review. • Laboratory assistant for the course of digital electronics design with DC-AC prototypes using the Arduino UNO board with an average score of 4.3/5 in the professor's review.	
Autonomous Surface Vessel - Backend Developer	01/2018 - 12/2018
Barranquilla, Atlántico	
• Developed a prototype related to Autonomous Surface Boats using a GPS receiver for measuring the position of the boat, a compass and other sensors and actuators for navigation using Python and Raspberry Pi platform. • Search algorithm using spherical geometry for the navigation with an error of 2 meters, leading to an acceptable error due to low-cost hardware limitations.	
AISMAP - Frontend Developer	01/2018 - 06/2018
Barranquilla, Atlántico	
• Build the UI for a mobile application on the Automatic Identification System (AIS) used by ships and vessel traffic services. • Accomplished time latency for the prototype close to 5s for transmitting data in real-time. • The AIVDM / AIVDO communication protocol was used to collect and integrate information from publicly available sources and vessels using the AIS protocol.	
Trucking Tracking - Full-Stack Developer	01/2018 - 06/2018
Barranquilla, Atlántico	
• Designed the UI for the trucks in transit with historical and real-time locations with an average response of 200ms using Javascript and HTML5 and CSS3. • Deployed the web application using Amazon Elastic Cloud Computing service EC2 for the server and the Amazon Relational Database Service for storage.	

PUBLICATIONS

International Journal on Communications Antenna and Propagation (IRECAP)	06/2019 - 12/2019
Obstacles, Speed and Spreading Factor: Insights in LoRa Mobile Performance.	

COURSES &
CERTIFICATES

Astronomer Certification DAG Authoring for Apache Airflow 3	10/2025
• DAG and DAG authoring • ETL • Apache Airflow.	
Astronomer Certification for Apache Airflow 3 Fundamentals	10/2025
• Data pipelines • Orchestation and scheduling • Apache Airflow and Dags	
Introduction to Kubernetes	02/2025
• Microservices • Minikube • Volume Management • ConfigMaps • Container Orchestation • Secrets • Ingress	
Google Project Management Professional Certificate	01/2025
• Agile project management • Budgeting and procurement • Risk management • Effective Communication • Project scoping and planning • Stakeholder management • Work management tools • Process improvement	
API Academy Certification Program	08/2020 - 05/2021
• API Designer • API Security Architect • API Product Manager	
Convolutional Networks for Visual Recognition	06/2018 - 08/2018
• CS231n.	
International Test of English Proficiency ITEP	04/2018 - 06/2018
• CEFR Level B2.	

EDUCATION

MSC on Electronics Engineering	06/2019 - 04/2023
Universidad del Norte	
Identification of expressive descriptors for style extraction in music analysis using linear and non linear models.	
BSC on Electronics Engineering	06/2014 - 01/2019
Universidad del Norte	
System for the measurement of musical similarity, using expressive markers considering the acoustic intensity and temporal metrics.	